



Universität Hamburg
DER FORSCHUNG | DER LEHRE | DER BILDUNG

Master Thesis

The impact of tropospheric versus stratospheric CO₂ forcing on the Hadley cell

Ole Merkes

Atmospheric Science (M.Sc.)
Matr.-Nr.: 7830665

First Supervisor: Prof. Dr. Sarah Kang
Second Supervisor: Dr. Moritz Günther

The impact of tropospheric versus stratospheric CO₂ forcing on the Hadley cell

Master Thesis

by

Ole Merkes

Student Name	Student Number
--------------	----------------

Ole Merkes	7830665
------------	---------

Supervisors: Prof. Dr. Sarah Kang
Dr. Moritz Günther



Universität Hamburg
DER FORSCHUNG | DER LEHRE | DER BILDUNG

Abstract

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, libero. Vivamus viverra fermentum felis. Donec nonummy pellentesque ante. Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem. Sed lacinia nulla vitae enim. Pellentesque tincidunt purus vel magna. Integer non enim. Praesent euismod nunc eu purus. Donec bibendum quam in tellus. Nullam cursus pulvinar lectus. Donec et mi. Nam vulputate metus eu enim. Vestibulum pellentesque felis eu massa.

Quisque ullamcorper placerat ipsum. Cras nibh. Morbi vel justo vitae lacus tincidunt ultrices. Lorem ipsum dolor sit amet, consectetur adipiscing elit. In hac habitasse platea dictumst. Integer tempus convallis augue. Etiam facilisis. Nunc elementum fermentum wisi. Aenean placerat. Ut imperdiet, enim sed gravida sollicitudin, felis odio placerat quam, ac pulvinar elit purus eget enim. Nunc vitae tortor. Proin tempus nibh sit amet nisl. Vivamus quis tortor vitae risus porta vehicula.

Contents

Summary	i
Nomenclature	iii
1 Introduction	1
2 Theory	2
2.1 The atmosphere under greenhouse gas forcing	2
2.2 Hadley circulation strength under greenhouse gas forcing	2
2.3 The angular momentum budget	2
2.3.1 Axisymmetric theory	2
2.3.2 Eddy driven mean circulation	2
2.4 The energy budget of the atmosphere	2
2.5 Model setup	2
2.5.1 ECHAM5	2
2.5.2 konrad	2
3 Methods	3
3.1 ECHAM5	3
3.2 konrad	3
3.3 Decomposition in forcing factors	3
4 Results	4
4.1 The atmospheric response	4
4.2 The energetic framework	6
4.2.1 The forcing	6
5 Discussion	8
6 Conclusion	9
References	11

Nomenclature

Abbreviations

Abbreviation	Definition
USA	United States of America
EU	European Union

Variable Names

Symbol	Definition	Unit
α	greek alpha	[N]
β	greek beta	[J]

1

Introduction

Establishing of context: What is the HC?

The Hadley cell (HC) is the dominant meridional circulation in the tropics. The region at the equator is associated with rising motion and collocated with inter-tropical convergence zone (ITCZ). The dry regions of subsidence in the subtropics are tight to the descending branch of the HC. In the annual mean there are two cell, which extend from 0 to 30 ° N in the Northern hemisphere and equivalently 0 to 30 ° S in the Southern hemisphere. Throughout the annual cycle there are strong variations in the HC's extent and strength: As the maximum of insolation moves poleward, so does the ITCZ and the HCs, resulting in a strong winter and a weaker summer HC. This seasonal migration is related to the global monsoons.

The HC is responsible for the poleward transportation of angular momentum (and energy) and the equatorward transportation of moisture. Therefore, the HC has strong implications for life on this planet. Its extent crucially modulates global precipitation patterns (<empty citation>). [Precipitation changes in the Sahel in the historical period. Paleoclimate HC and precipitation changes.] (this is unequal to the significance)

Climate changes had a significant influence on the HC altering its strength and extent. This is why there has been a large effort to find out the implications of human driven greenhouse gas emissions in the current era.

Stating the Problem: To date, there has not been sufficient research that examines the connection between...

inter-model uncertainty

HC response depends on forcing structure This has been shown for quadrupling of CO₂ at different latitudes (Kim et al., 2022). Thus, we want to investigate how CO₂ forcing in the vertical alters atmospheric circulation and want to employ different frameworks for the HC dynamics. The HC response to this forcing is not what one would expect from the literature.

Scope Can we explain this using the common theories of HC dynamics.

Significance

advance knowledge in the applicable field; that is, it revises or creates new knowledge (for example, the results will extend what is known about the applicability of a theory, the results are widely generalizable, or, for qualitative studies, transferable to other contexts) - show applicability of different theories - transfers results to a different context **Overview over the study** Is this really necessary?

2

Theory

2.1. The atmosphere under greenhouse gas forcing

tropospheric warming
lapse rate feedback
stratospheric cooling

2.2. Hadley circulation strength under greenhouse gas forcing

Which drivers have been identified for the weakening of the HC

2.3. The angular momentum budget

Why is there a circulation in the first place

2.3.1. Axisymmetric theory

angular momentum from the equator is dissipated/transported to higher latitudes

2.3.2. Eddy driven mean circulation

Eddies can drive a circulation (Ferrel cell) or as in the case of the HC significantly enhance it

2.4. The energy budget of the atmosphere

diagnostic perspective: energy imbalance requires transport ocean, mmc. and eddies can transport energy

2.5. Model setup

2.5.1. ECHAM5

2.5.2. konrad

3

Methods

- 3.1. ECHAM5
- 3.2. konrad
- 3.3. Decomposition in forcing factors

4

Results

4.1. The atmospheric response

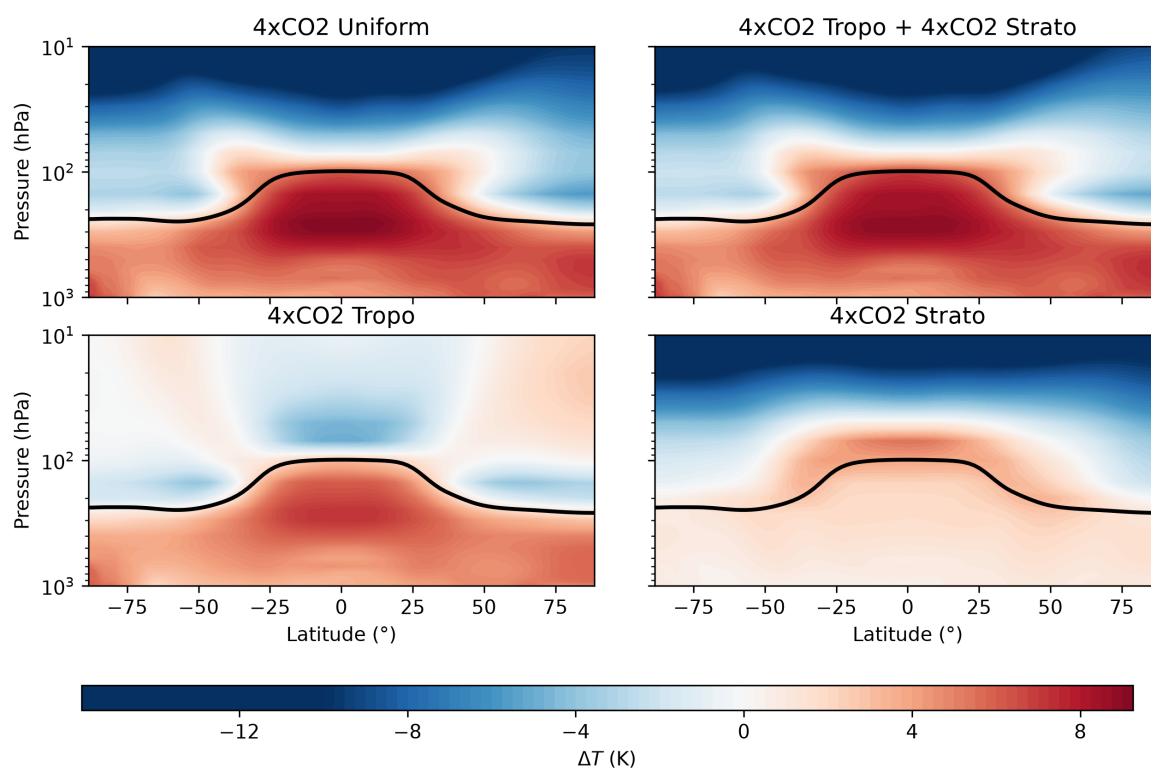


Figure 4.1

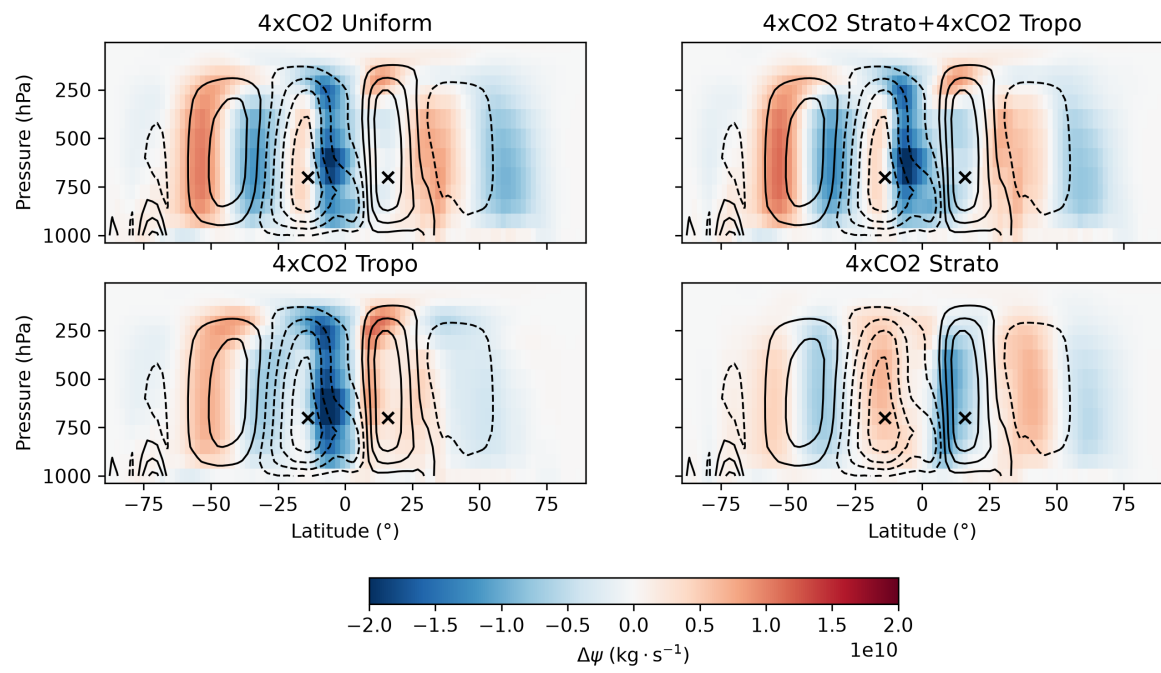


Figure 4.2

4.2. The energetic framework

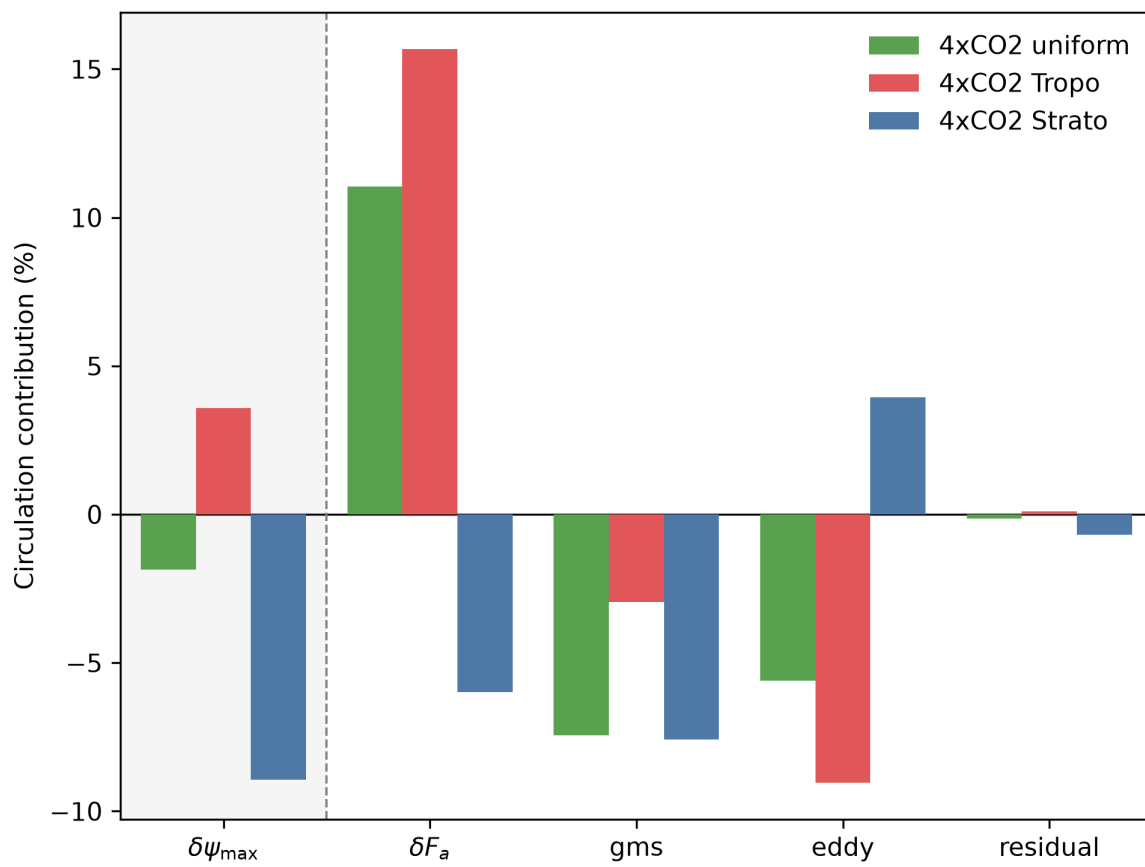


Figure 4.3

4.2.1. The forcing

Explain the forcing structure (Merlis, 2015)

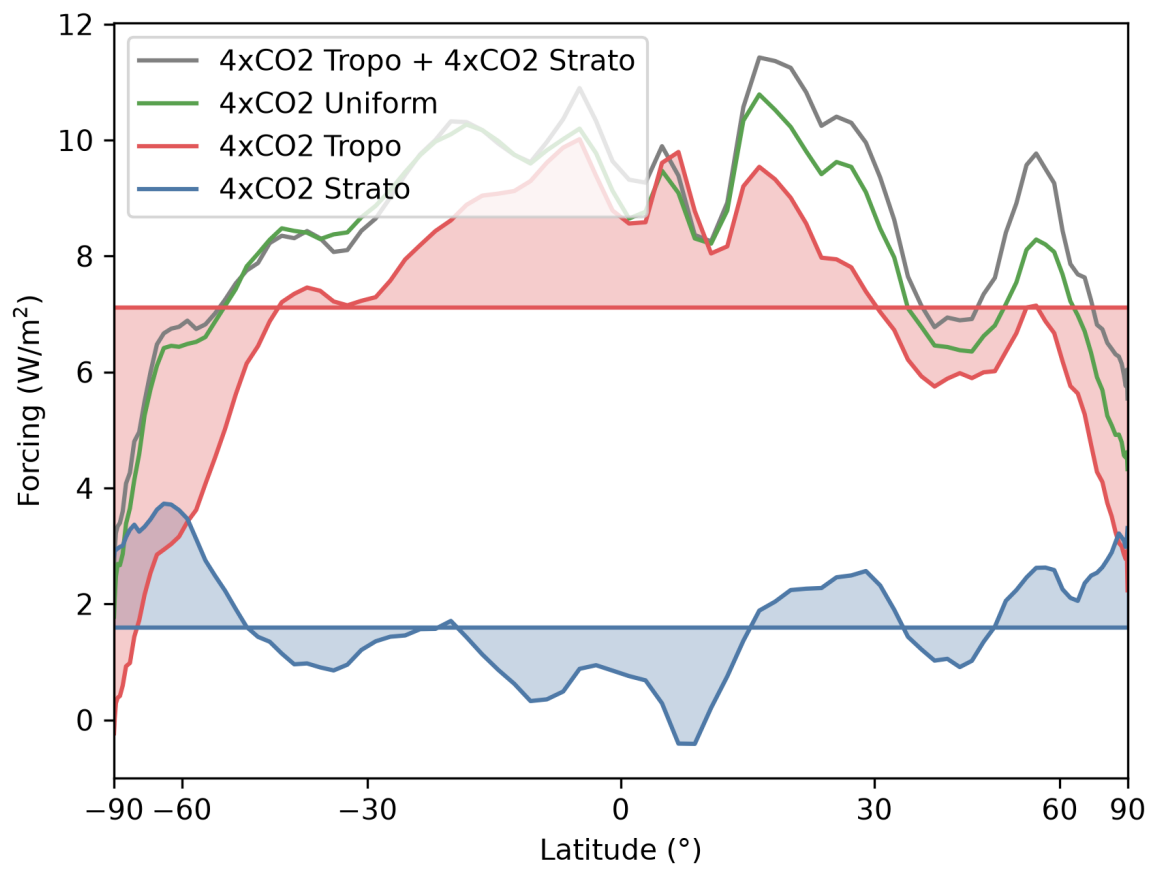


Figure 4.4

5

Discussion

6

Conclusion

Acknowledgement

First, I like to thank my first supervisor , for Also, I want to thank my second supervisor for

Thanks to for all the insights into Thanks to for all the intense but fruitful scientific debate about Thanks to ... , who always had an open door for

Also, I would like to thank the

And last but not least, thanks to , who helped As well as for the help with

References

- Kim, D., Kim, H., Kang, S. M., Stuecker, M. F., & Merlis, T. M. (2022). Weak hadley cell intensity changes due to compensating effects of tropical and extratropical radiative forcing. *npj Climate and Atmospheric Science*, 5(1), 61. <https://doi.org/10.1038/s41612-022-00287-x>
- Merlis, T. M. (2015). Direct weakening of tropical circulations from masked CO₂ radiative forcing. *Proceedings of the National Academy of Sciences*, 112(43), 13167–13171. <https://doi.org/10.1073/pnas.1508268112>

Eidesstattliche Versicherung

Hiermit versichere ich an Eides statt, dass ich die vorliegende Arbeit im Studiengang selbstständig verfasst und keine anderen als die angegebenen Hilfsmittel – insbesondere keine im Quellenverzeichnis nicht benannten Internet-Quellen – benutzt habe. Alle Stellen, die wörtlich oder sinngemäß aus Veröffentlichungen entnommen wurden, sind als solche kenntlich gemacht. Ich versichere weiterhin, dass ich die Arbeit vorher nicht in einem anderen Prüfungsverfahren eingereicht habe und die eingereichte schriftliche Fassung der auf dem elektronischen Speichermedium entspricht.

Einer Veröffentlichung der vorliegenden Arbeit in der zuständigen Fachbibliothek des Fachbereichs stimme ich zu.

Hamburg, den _____ Unterschrift: _____